

COMPUTER NETWORK PROJECT

Sarah bahswhan
Deema alsini



Network Traffic Monitoring Project

The Network Traffic Monitoring Project is a real-time tool that records, analyzes, and traces network traffic to give a comprehensive view of network performance. Use the Scapy library to effectively track and analyze packets and extract data, including source and destination addresses, protocol types, and packet sizes. Throughput and latency are used to identify problems with network performance and behavior. For scalability and real-time processing, the system also has a multithreaded TCP server and Matplotlib for display it . It is important technology for network managers to efficiently monitor their networks

b) Project Outline

Packages:

- Import all necessary libraries:
- Scapy for network packet analysis.
- Threading for multi-threaded execution.
- Logging for storing network events.
- Matplotlib for visualizing network data.

Data Structures: Set up data structures to track packet sizes, throughput, latency, and unique addresses.

Functions: Implement functions for logging events, updating data, processing packets, generating graphs, and monitoring the network.

Main: Start packet sniffing and monitoring using separate threads.

the main method to collects and processes network data :

Packet Sniffing and Analysis: Scapy captures Ethernet, IP, TCP, and UDP packets, extracting information like source and destination addresses and packet size.

Real-Time Data Processing: Compute metrics like throughput (bits per second) and latency (milliseconds) at regular intervals.

Data Visualization:

- Use Matplotlib to create graphs for:
- Throughput over time.
- Latency distribution.
- Protocol usage.

Log Management: Record packet details such as protocol type, timestamps, size, and source/destination addresses in `network_events.log`.

Processing Logged Data: Analyze the logged data to identify patterns and trends, and use this information to generate visualizations and compile network performance metrics.

c) Functions Used

Some of the significant things done for the project are:-

og_event(protocol, src_addr, dest_addr, message_size, timestamp): Logs timing information and packet details (protocol type, source/destination addresses, packet size) into a log file.

update_event_data(protocol, src_addr, dest_addr, message_size, timestamp): Updates metrics such as throughput, packet counts, and unique IP or MAC addresses.

process_packet(packet): Processes individual packets, extracting necessary information for analysis.

calculate_throughput(interval): *Calculates the throughput of each protocol over a specified time interval.*

calculate_latency(): Measures the round-trip latency for live TCP/UDP connections.

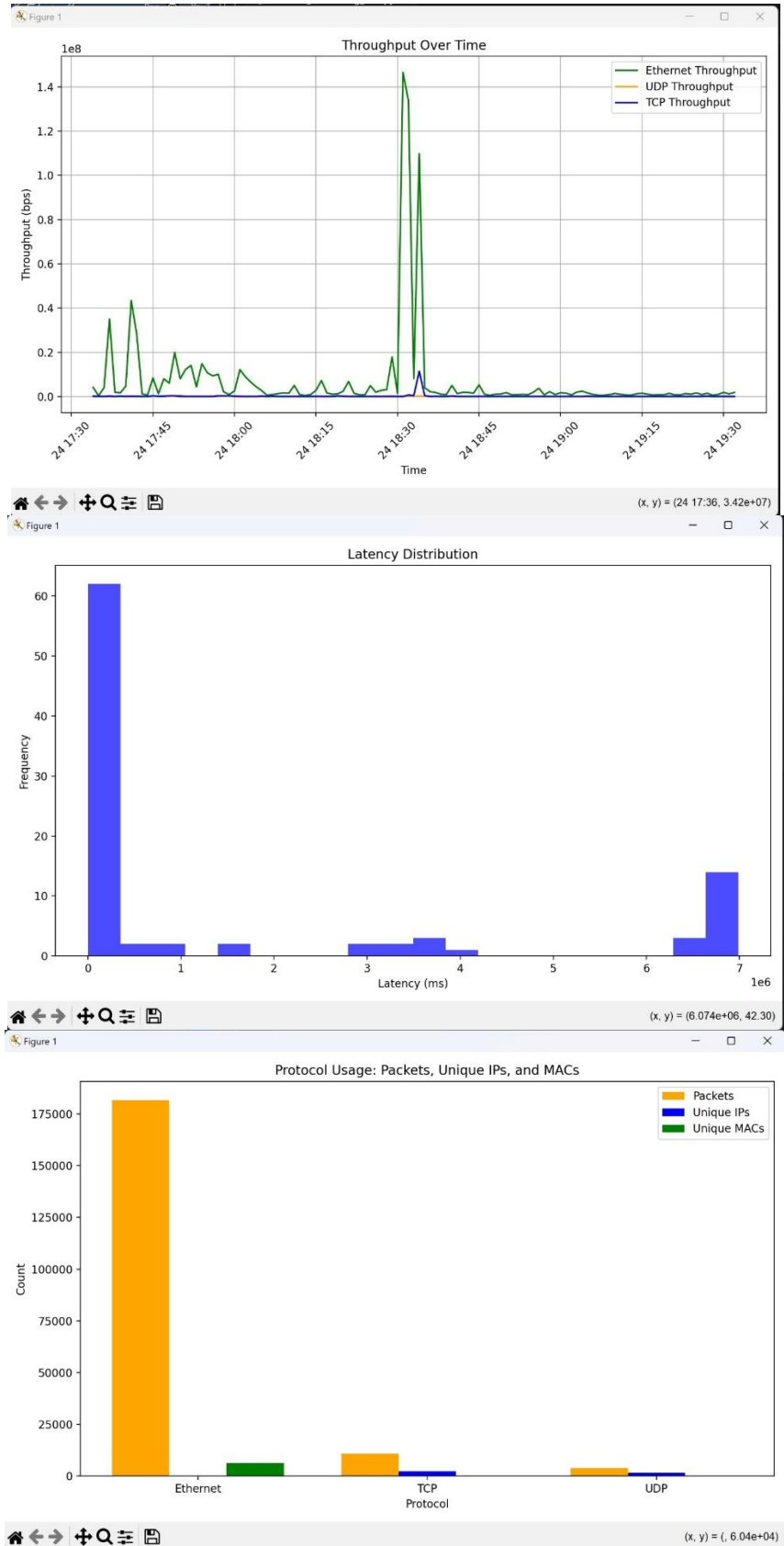
plot_graphs(): Generates graphs for throughput, latency distribution, and protocol usage using logged data.

monitor_loop(): Continuously monitors the network, updating statistics and metrics in real time

Graphical Analysis

The following visualizations were built using the network data that we obtained:

Screenshots :



Include relevant screenshots of:

- Console outputs during real-time monitoring.

```
1 # Import packages
2 from scapy.all import sniff, Ether, IP, TCP, UDP
3 import threading
4 import time
5 import datetime
6 from collections import defaultdict
7 import matplotlib.pyplot as plt
8
9 # Data structures needed for tracking data
10
11 Unique IP Addresses: 47
12 Unique MAC Addresses: 9
13
14 --- Throughput (bps) ---
15 Ethernet: 11747.20 bps
16 TCP: 217.60 bps
17 UDP: 297.60 bps
18 Average Latency: 1618152.74 ms
19
20 --- Network Statistics ---
21 Ethernet Connections: 175323, Average Size: 590.11 bytes
22 TCP Connections: 10299, Average Size: 242.53 bytes
23 UDP Connections: 3592, Average Size: 299.98 bytes
24 Rate of New Connections (Ethernet) (last 30 seconds): 191 connections
25 Rate of New Connections (TCP) (last 30 seconds): 0 connections
26 Rate of New Connections (UDP) (last 30 seconds): 0 connections
27 Unique IP Addresses: 47
28 Unique MAC Addresses: 9
29
30 --- Throughput (bps) ---
31 Ethernet: 6881.60 bps
32 TCP: 761.60 bps
33 UDP: 99.20 bps
34 Average Latency: 1620679.25 ms
35
36 --- Network Statistics ---
37 Ethernet Connections: 175674, Average Size: 589.63 bytes
38 TCP Connections: 10319, Average Size: 242.23 bytes
39 UDP Connections: 3603, Average Size: 299.59 bytes
40 Rate of New Connections (Ethernet) (last 30 seconds): 288 connections
41 Rate of New Connections (TCP) (last 30 seconds): 0 connections
42 Rate of New Connections (UDP) (last 30 seconds): 0 connections
43 Unique IP Addresses: 47
44 Unique MAC Addresses: 9
45
46 --- Throughput (bps) ---
47 Ethernet: 5606.40 bps
48 TCP: 217.60 bps
49 UDP: 80.00 bps
50 Average Latency: 1631144.24 ms
51
52 --- Network Statistics ---
53 Ethernet Connections: 175674, Average Size: 589.63 bytes
54 TCP Connections: 10319, Average Size: 242.23 bytes
55 UDP Connections: 3603, Average Size: 299.59 bytes
56 Rate of New Connections (Ethernet) (last 30 seconds): 288 connections
57 Rate of New Connections (TCP) (last 30 seconds): 0 connections
58 Rate of New Connections (UDP) (last 30 seconds): 0 connections
59 Unique IP Addresses: 47
60 Unique MAC Addresses: 9
61
62 --- Throughput (bps) ---
63 Ethernet: 22075.20 bps
64 TCP: 548.00 bps
65 UDP: 1072.00 bps
66 Average Latency: 1633799.07 ms
67
68 --- Network Statistics ---
69 Ethernet Connections: 175788, Average Size: 589.41 bytes
70 TCP Connections: 10327, Average Size: 242.11 bytes
71 UDP Connections: 3610, Average Size: 299.38 bytes
72 Rate of New Connections (Ethernet) (last 30 seconds): 203 connections
```

```
network_events.log
File Edit View

2024-11-24 17:34:27 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 438 bytes
2024-11-24 17:34:27 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 74 bytes
2024-11-24 17:34:27 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 257 bytes
2024-11-24 17:34:27 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 139 bytes
2024-11-24 17:34:27 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 1284 bytes
2024-11-24 17:34:27 - Ethernet - Source: 4c:ba:d7:de:20:1c, Destination: 01:00:5e:7f:ff:fa - Size: 236 bytes
2024-11-24 17:34:27 - UDP - Source: 192.168.8.6, Destination: 239.255.255.250 - Size: 236 bytes
2024-11-24 17:34:27 - Ethernet - Source: 4c:ba:d7:de:20:1c, Destination: 01:00:5e:00:00:fb - Size: 88 bytes
2024-11-24 17:34:27 - UDP - Source: 192.168.8.6, Destination: 224.0.0.251 - Size: 88 bytes
2024-11-24 17:34:28 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 187 bytes
2024-11-24 17:34:28 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 74 bytes
2024-11-24 17:34:28 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 116 bytes
2024-11-24 17:34:28 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 540 bytes
2024-11-24 17:34:28 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 74 bytes
2024-11-24 17:34:28 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 139 bytes
2024-11-24 17:34:28 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 1335 bytes
2024-11-24 17:34:28 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 116 bytes
2024-11-24 17:34:28 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 539 bytes
2024-11-24 17:34:28 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 74 bytes
2024-11-24 17:34:29 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 74 bytes
2024-11-24 17:34:29 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 74 bytes
2024-11-24 17:34:30 - Ethernet - Source: 2e:74:da:a0:f0:2d, Destination: 28:c5:d2:fe:e5:bb - Size: 54 bytes
2024-11-24 17:34:30 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 94 bytes
2024-11-24 17:34:30 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 94 bytes
2024-11-24 17:34:30 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 94 bytes
2024-11-24 17:34:30 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 184 bytes
2024-11-24 17:34:30 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 172 bytes
2024-11-24 17:34:30 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 108 bytes
2024-11-24 17:34:30 - TCP - Source: 192.168.8.57, Destination: 3.65.102.105 - Size: 108 bytes
2024-11-24 17:34:30 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 86 bytes
2024-11-24 17:34:30 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 94 bytes
2024-11-24 17:34:31 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 86 bytes
2024-11-24 17:34:31 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 74 bytes
2024-11-24 17:34:31 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 2074 bytes
2024-11-24 17:34:31 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 257 bytes
2024-11-24 17:34:31 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 54 bytes
2024-11-24 17:34:31 - TCP - Source: 3.65.102.105, Destination: 192.168.8.57 - Size: 54 bytes
2024-11-24 17:34:31 - Ethernet - Source: 02:17:6a:42:c2:70, Destination: 28:c5:d2:fe:e5:bb - Size: 110 bytes
2024-11-24 17:34:31 - TCP - Source: 3.65.102.105, Destination: 192.168.8.57 - Size: 110 bytes
2024-11-24 17:34:31 - Ethernet - Source: 2e:74:da:a0:f0:2d, Destination: 28:c5:d2:fe:e5:bb - Size: 144 bytes
2024-11-24 17:34:31 - Ethernet - Source: 28:c5:d2:fe:e5:bb, Destination: 02:17:6a:42:c2:70 - Size: 54 bytes
2024-11-24 17:34:31 - TCP - Source: 192.168.8.57, Destination: 3.65.102.105 - Size: 54 bytes
2024-11-24 17:34:31 - Ethernet - Source: 4c:ba:d7:de:20:1c, Destination: 28:c5:d2:fe:e5:bb - Size: 66 bytes
Ln 1, Col 1 21,165,492 characters 100% Windows (CRLF) UTF-8
```

e) Discussion

Latency Distribution:

Most connections are very low latency as shown by the histogram, which peaks around 60 ms. In other words, a long(dof)tail means that some connections observed much higher latency.

Throughput Over Time:

Throughput for Ethernet, TCP and UDP is shown in the line graph. Networking traffic is clearly dominated by Ethernet which exhibits the highest and a constant throughput. TCP and UDP ramp up a little but not as high as Ethernet

Protocol Usage:

It is a bar chart that gives us the counts of packets for each protocol along with unique IPs and MACs. Ethernet takes the top in all measurements, underscoring its primary significance in the system. TCP and UDP traffic exists, but on a very minor level (a sign it is being not utilized).

Challenges Noted:

1. Multi-threading: We had a multiplicity of threads running, that needed precise synchronizing to avoid race conditions and corruption of state when reading/writing shared resources. It was also relatively resource-heavy, and had good error handling.
2. Scapy Installation and Configuration — Scapy installation demanded elevated permissions, weighed compatibility concerns across disparate OSes and Python versions.
3. Merging Graphs: Custom and optimized to handle performance issues, different graph types aligned over one layout with real-time updates and interactivity.